Analysis of Temperatures Trends for cities across The Netherlands

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Introduction

Throughout the twentieth century, the Netherlands, similarly to the rest of the world, underwent a significant change in the way people and goods are transported. From the widespread adoption of private automobiles to the rapid growth of Schiphol Airport and the expansion of the national road network. The country experienced great economic success. However, the carbon footprint also rose dramatically. This paper investigates whether these developments left a measurable impact on the Dutch climate, using temperature data recorded across three cities: Maastricht, Eelde, and De Bilt.

The dataset spans from 1907 to 2024 and contains the annual average temperature alongside the smoothed monthly averages for all three locations. To structure our analysis, we split the dataset into two periods that show distinct phases of automobile and aviation development. The first period, 1907-1960, reflects the early and mainly untouched atmosphere. Car ownership and the national road network were limited, and commercial aviation was just at its start. The second period, 1961-2024, captures the time of mass automobile and aviation expansion. Commercial car ownership escalated from fewer than one million registered in the 1960s to over nine million by the 2020s. At the same time, commercial aviation has greatly developed, with Amsterdam Schiphol becoming one of the busiest European airports. Importantly, these developments greatly increased greenhouse gas emissions in the Netherlands.

The research question guiding this paper is therefore:

Can different inferential statistics tools identify a linear upward trend in temperature across three Dutch cities over the period 1907–2024, and is there statistical evidence that this trend accelerated in the post-1960 period, coinciding with the rise of mass automobile and aviation use in the Netherlands?

The main hypothesis is that temperatures show a positive trend over time in all locations and that this trend has strengthened in the second half of the considered period. To answer this question, the analysis is conducted in several stages. First, average temperatures over various time intervals are compared using both overlapping and non-overlapping averages to identify potential changes in temperature levels. Second, linear regression models are estimated to quantify the relationship between time and temperature and to test for the presence of a statistically significant trend. The data is then divided into subperiods to test whether this trend has accelerated over time. Finally, bootstrap methods are applied to assess the reliability of the results without relying solely on standard parametric assumptions. Based on all the methods described above, this study aims to provide a comprehensive overview of how temperatures are rising in various regions of the Netherlands and whether there is evidence of an upward trend in warming over time.

Comparing Average Temperatures Across Periods

To obtain an initial understanding of temperature developments, average temperatures are compared across different time periods using both rolling averages and non-overlapping block averages. This approach allows us to identify long-term patterns while reducing the impact of short-term fluctuations.

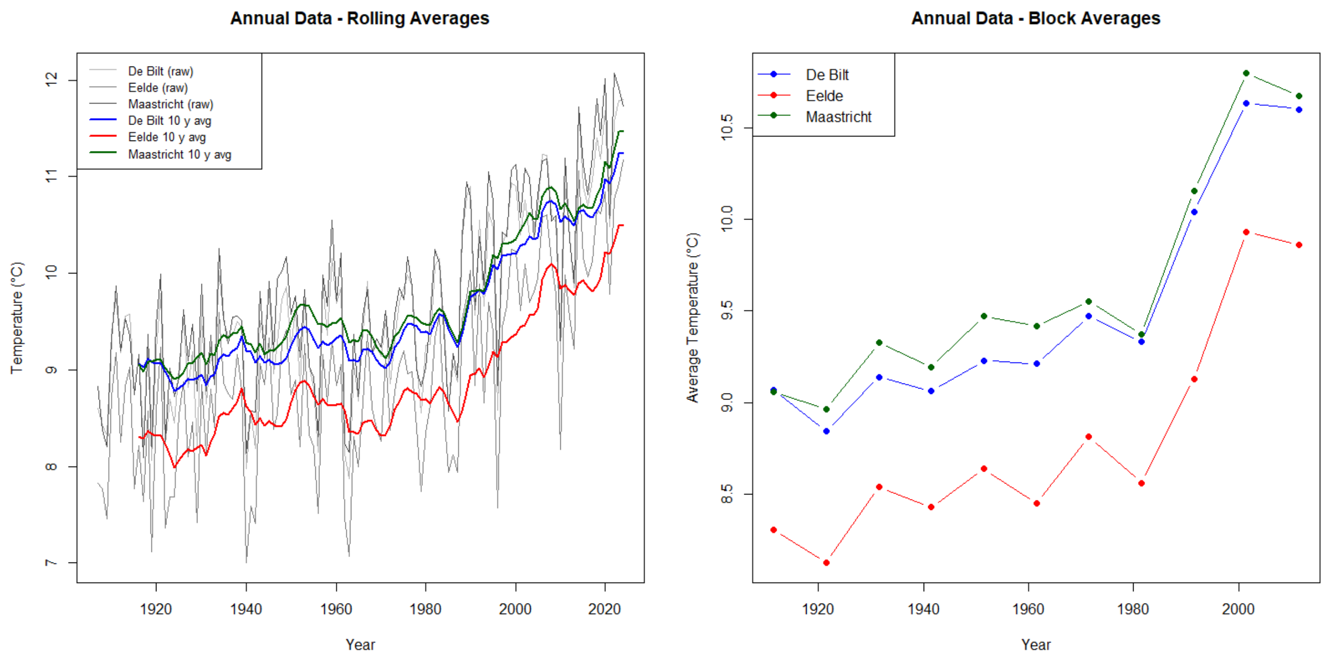

Figure 1: Annual Graphs depicting overlapping and non-overlapping averages

The left panel of the figure shows 10-year rolling averages for the cities of De Bilt, Elde, and Maastricht. The smoothed data series reveal a gradual increase in temperatures over time. Despite fluctuations in the early decades, a clear upward trend is observed in all three locations starting around the 1980s.

The right panel presents non-overlapping 10-year averages, which provide a clearer comparison across distinct time intervals. In the early part of the sample (approximately 1910–1970), average temperatures remain relatively stable, with only modest variation across decades. Although some gradual increase is visible, the changes are limited in magnitude.

A noticeable shift has been observed since around the 1980s. Average temperatures have risen significantly in all three locations. This increase was particularly pronounced between the 1980s and the 2000s, when all cities experienced a sharp spike in average temperatures. Maastricht consistently records the highest temperatures, followed by De Bilt, while Elde remains the coolest location throughout the entire observation period. Despite these differences in levels, the overall pattern of rising temperatures is similar across all regions.

Comparing early and late periods suggests that the increase in temperature is not smooth over time. Instead, the data indicate that warming accelerates in the later decades of the sample. This observation provides preliminary evidence supporting the hypothesis of a positive and increasing temperature trend over time.

Monthly data extension

The same procedures were also applied to the smoothed monthly data to compare the average values.

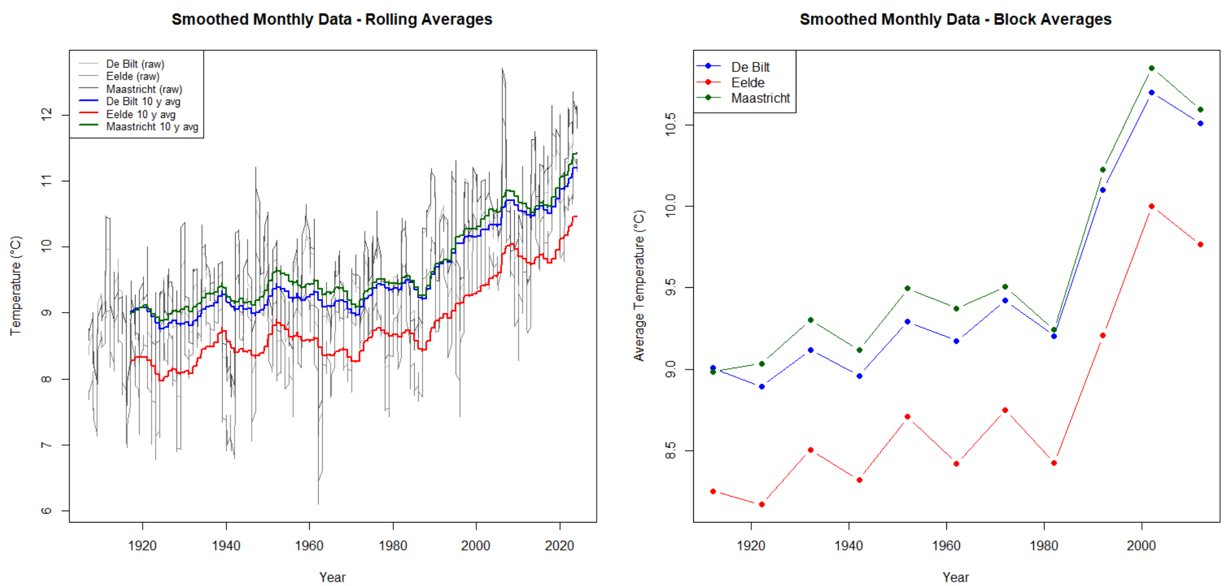

Figure 2: Smoothed Monthly Graphs depicting overlapping and non-overlapping averages

The analysis of smoothed monthly averages further supports the previous findings. As shown in Figure 2, both the rolling averages and block averages display a clear upward trend over time for all cities. Compared to earlier periods, average temperatures increase noticeably around 1980. The smoothed series highlights this pattern more clearly by reducing short-term fluctuations, confirming that the warming trend is persistent and not driven by temporary variations.

Assumptions

Before relying on the results of the linear regression model, it is necessary to verify whether the standard assumptions underlying OLS are reasonably satisfied. This is done by analyzing the residuals obtained from the regression of temperature on time. Since the patterns are very similar across all locations, we focus on Maastricht, while noting that the conclusions apply equally to De Bilt and Eelde.

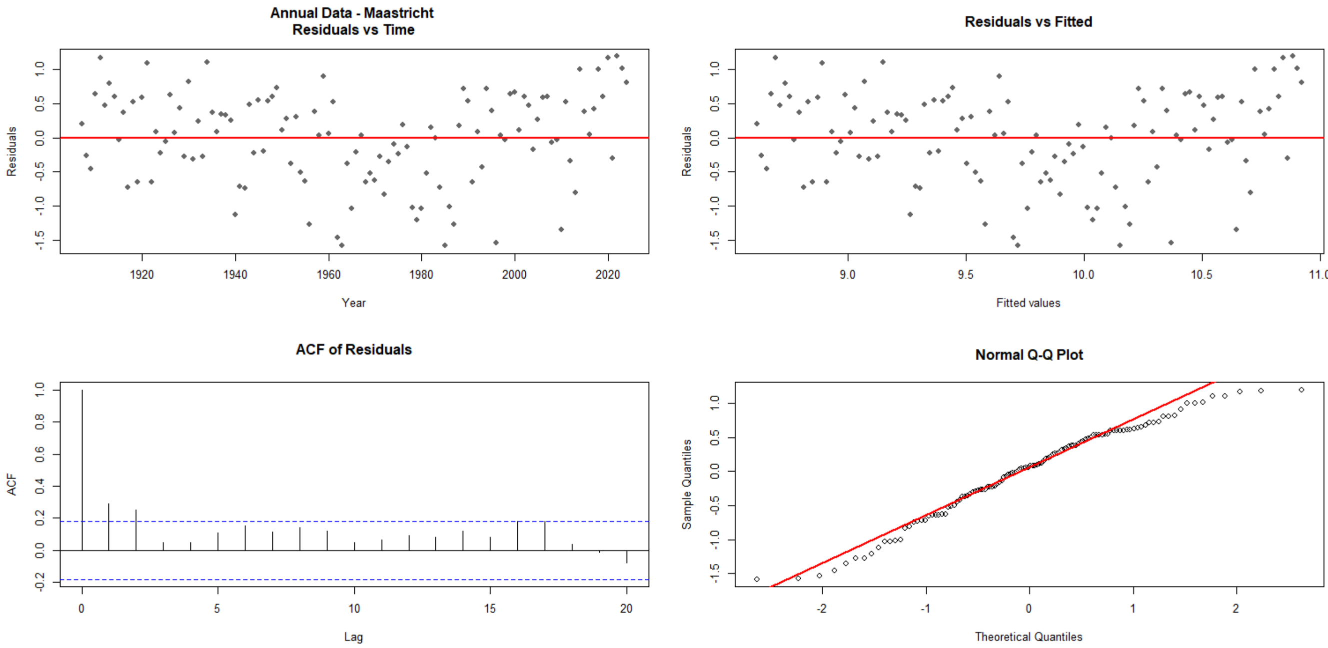

Figure 3: Annual data assumptions test, Maastricht

First, we examine whether a linear relationship between temperature and time is appropriate. This is assessed by plotting the residuals against time. The residuals are distributed around zero without any noticeable bias or systematic pattern. This suggests that the linear specification provides a reasonable approximation of the underlying relationship, and there is no strong evidence of non-linearity in the data.

Next, we consider whether the variance of the error term remains constant over time. This is evaluated using a plot of residuals against fitted values. The distribution of residuals appears relatively stable across the entire range of estimated values, with no clear bell-shaped curve or expansion of the distribution observed. This indicates that the variance of the errors remains approximately constant, which supports the assumption of homoscedasticity.

We then investigate whether the residuals are independent over time. This is done using the sample autocorrelation function (ACF) of the residuals. For Maastricht, the first-order autocorrelation is approximately 0.29, which exceeds the approximate significance bound of ±0.18. This indicates the existence of positive serial correlation in the residuals. In practice, this means that temperature variations tend to persist over time, so neighboring observations are not entirely independent. As a result, the standard errors obtained from ordinary least squares (OLS) regression may be underestimated, leading to an overestimation of statistical significance.

Finally, we assess whether the residuals follow a normal distribution. This is examined using a normal Q–Q plot. The points lie close to the reference line, with only minor deviations in the tails. This is further supported by a high Q–Q correlation coefficient (approximately 0.99), indicating that the residuals are very close to normally distributed. Given the relatively large sample size, any small deviations from normality are not a major concern.

Overall, the assumption checks suggest that the linear model is appropriate in terms of functional form and variance stability, and the residuals are approximately normally distributed. The main limitation is the existence of serial correlation, which may affect the validity of standard conclusions. This issue is addressed in the following section using bootstrap methods, which provide more robust results in the context of dependence.

Monthly extension

We repeat the same diagnostic procedure for the smoothed monthly data. As before, we focus on Maastricht, noting that the results are nearly identical across all cities.

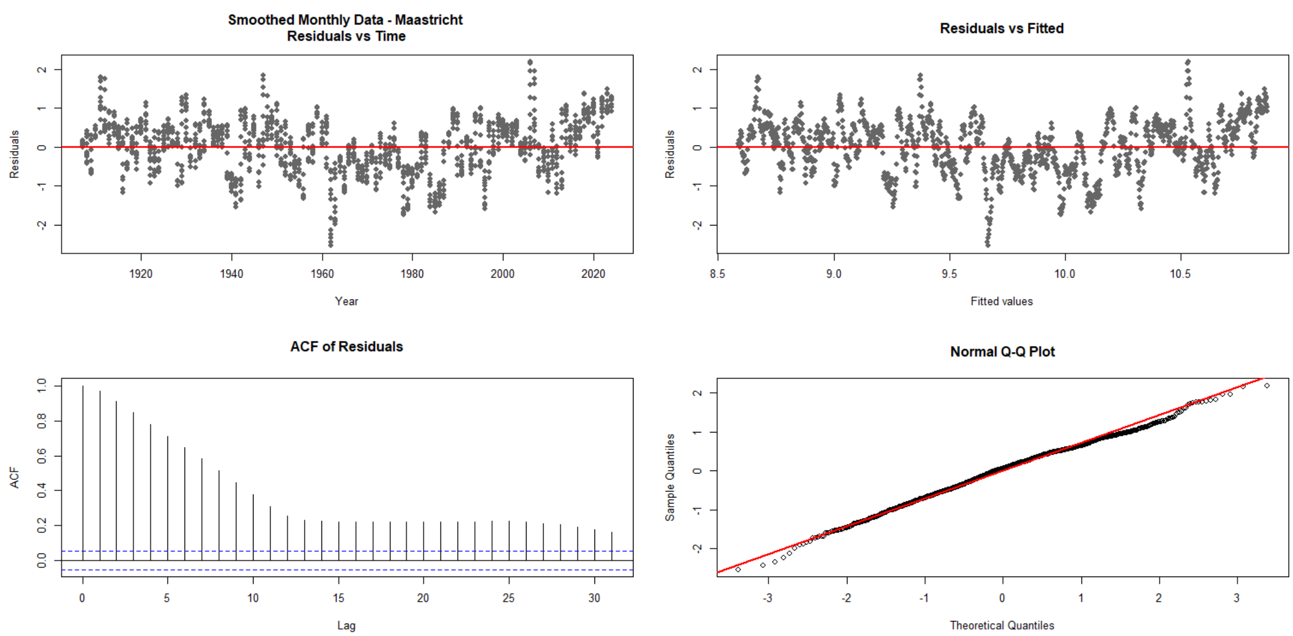

Figure 4: Annual data assumptions test, Maastricht

The visual inspection of residuals against time again shows that they fluctuate around zero without any clear non-linear pattern. This supports the use of a linear specification. Similarly, the residuals plotted against fitted values do not exhibit a clear change in spread, suggesting that the variance remains relatively stable.

The main difference compared to the annual data arises in the dependence structure of the residuals. The autocorrelation function demonstrates exceptional durability. For Maastricht, the first-order autocorrelation is approximately 0.97, which is far above the corresponding significance bound of ±0.053. Moreover, the ACF decays very slowly across higher lags, indicating that dependence extends over long periods of time.

This result is not surprising given the nature of the data. Monthly temperatures are inherently highly persistent, especially after smoothing, as each observation is strongly related to neighbouring months. As a result, the effective amount of independent information in the sample is much smaller than the number of observations suggests, and standard OLS inference is likely to be overly optimistic.

Finally, the normality assumption appears to be well satisfied. The Q–Q plot closely follows the reference line, and the correlation between empirical and theoretical quantiles is very high (approximately 0.99). Given the large sample size, this confirms that the distribution of residuals is sufficiently close to normal.

Overall, the conclusions are similar to the annual case, but with a much stronger violation of independence. While the linear specification and variance assumptions remain reasonable, the pronounced serial correlation in the monthly data reinforces the need for bootstrap-based inference, which is more robust to such dependence.

Bootstrap necessity

The existence of serial correlation in both datasets, and especially in the smoothed monthly data, has important implications for statistical inference. Since OLS assumes independent observations, this violation leads to underestimated standard errors and overly optimistic t-statistics and p-values. As a result, the usual parametric results may not be reliable.

To address this issue, in the next section we supplement our analysis with bootstrap methods. Bootstrap estimates are not based on such strict assumptions of independence and yield more robust estimates of the sampling distribution of the regression coefficients. This makes them particularly suitable when there are dependency structures, such as those observed in the temperature data.

Regression Model

To formally examine the existence of a temperature trend over time, we estimate a linear regression model relating temperature to time. The model is specified as:

$$T_t = \alpha + \beta t + \varepsilon_t,$$

where T_t denotes temperature at time t, and βcaptures the trend in temperature over time.

The analysis is performed separately for each location (De Bilt, Eelde, and Maastricht). For the annual dataset, each observation corresponds to one year, meaning that the slope coefficient βrepresents the average yearly change in temperature.

To assess whether temperatures exhibit an upward trend, we test the following hypotheses:

$$H_0: \beta = 0 \text{ against } H_1: \beta \neq 0$$

The null hypothesis implies that there is no trend in temperature over time, while the alternative hypothesis suggests a positive trend.

Results — Annual Data

City	Intercept (α)	Slope (β, °C/year)	SE(β)	t-statistic	p-value	95% CI for β
De Bilt	8.4853	0.0194	0.0018	10.9318	< 2.22e-16	[0.0159 , 0.0229]
Eelde	7.8220	0.0185	0.0019	9.6543	< 2.22e-16	[0.0147 , 0.0223]
Maastricht	8.5948	0.0197	0.0018	10.8549	< 2.22e-16	[0.0161 , 0.0233]

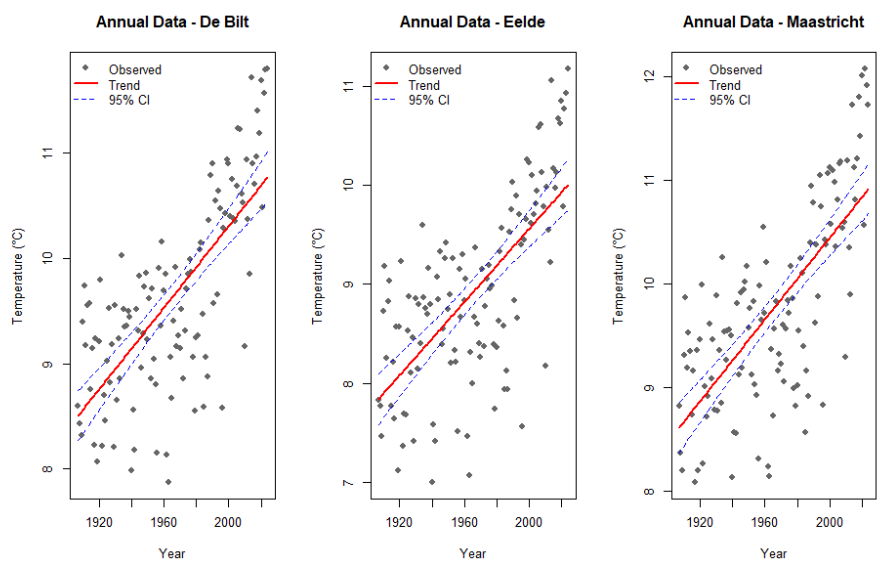

Figure 5: Regression graphs annual data

The results show a clear and consistent positive relationship between temperature and time across all locations. The estimated slope coefficients range between approximately 0.0185 and 0.0197 °C per year, indicating that temperatures have increased steadily over the observed period.

The corresponding t-statistics are large in magnitude, and all p-values are effectively zero, leading to a strong rejection of the null hypothesis $H_0: \beta = 0$. In addition, the 95% confidence intervals for all slope coefficients lie strictly above zero, further confirming the statistical significance of the estimated trends.

Importantly, the similarity of the estimated slopes across the three cities suggests that the warming pattern is not location-specific but reflects a broader regional trend. While individual observations exhibit some variability around the fitted lines, the overall upward trajectory is clearly visible in all cases.

Monthly extension

Regression Results for Smoothed Monthly Data

City	Intercept (α)	Slope (β, °C/month)	SE(β)	t-statistic	p-value	95% CI for β
De Bilt	8.4777	0.0016	0.0000	35.8994	< 2.22e-16	[0.0015 , 0.0017]
Eelde	7.8139	0.0015	0.0000	31.6943	< 2.22e-16	[0.0014 , 0.0016]
Maastricht	8.5861	0.0016	0.0000	35.9053	< 2.22e-16	[0.0015 , 0.0017]

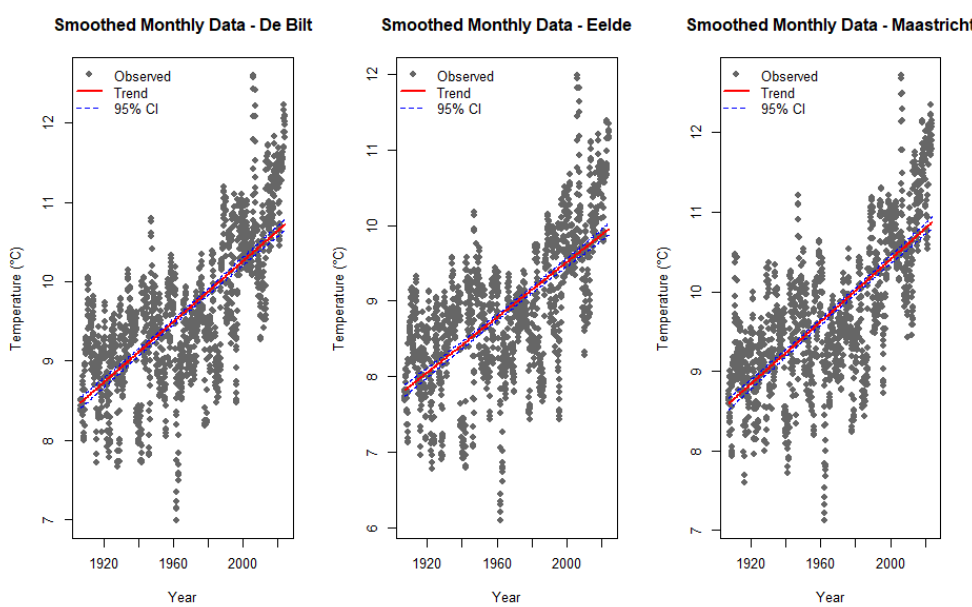

Figure 6: Regression graphs for smoothed monthly data

The regression results for the smoothed monthly data (see Figure 6) confirm the same positive trend observed in the annual data, but with substantially higher precision.

Compared to the annual data, the estimated slope coefficients for the smoothed monthly data are numerically much smaller (around 0.0015–0.0016). However, this difference is purely due to the change in time scale. In the monthly dataset, each observation represents one month rather than one year, so the slope measures the temperature change per month instead of per year.

When adjusting for this difference (multiplying by 12), the monthly estimates correspond closely to the annual slopes (approximately 0.018–0.019°C per year). This shows that the underlying trend is consistent across datasets, despite the difference in scale.

In addition, the monthly data produce much larger t-statistics and narrower confidence intervals. This reflects the larger number of observations, which increases estimation accuracy. The reported standard errors for the smoothed monthly data appear as zero due to rounding. In reality, they are very small but positive, reflecting the high precision of the estimates resulting from the large number of observations.

Overall, the null hypothesis $H_0: \beta = 0$ is strongly rejected for all cities. The results confirm that the upward temperature trend is both statistically significant and robust across different data frequencies.

Regression Conclusion

The regression analysis shows that the estimated slope coefficients are positive for all cities and across both annual and smoothed monthly datasets. This provides clear evidence of an upward trend in temperatures over time. For instance, in the case of Maastricht's annual data, the estimated slope indicates that temperature increases on average by approximately 0.0197°C per year.

In addition, the results are highly statistically significant, with large test statistics and very small p-values, allowing us to reject the null hypothesis of no trend. The consistency of these findings across all locations further suggests that the observed warming pattern is not driven by local factors, but reflects a broader regional development.

These results align with the historical context discussed earlier in the report, where increased industrial activity and the expansion of transportation, including the rise of the automobile and aviation sectors, are often associated with higher emissions and environmental changes. While the regression analysis itself does not establish causality, it provides strong statistical evidence of a persistent increase in temperatures that is consistent with these broader historical developments.

Samples Regression

Choice of Breakpoint

To further investigate potential changes in the temperature trend over time, the sample is divided into two subperiods using 1960 as a breakpoint. This choice is motivated both by visual inspection of the data and by historical considerations.

From the graphical analysis of averages, a more notable increase in temperatures becomes visible around the late 1970s to early 1980s. However, selecting this point directly as a breakpoint could bias the analysis, as it would isolate only the period of strongest increase and potentially overstate the difference between subsamples.

Instead, the year 1960 is chosen as a more neutral and conservative division. This allows the later period to capture the full development of the stronger warming trend, including the acceleration observed in the subsequent decades, while avoiding a breakpoint that is determined solely by the most extreme part of the data.

In addition, the period after 1960 aligns with important historical developments, such as increased industrial activity and the expansion of transportation, which are often associated with rising emissions. Therefore, this division provides a balanced framework to compare an earlier period with more moderate changes to a later period characterized by sustained and intensifying warming.

Hypothesis

To examine whether the temperature trend has changed over time, the sample is divided into two periods (1907–1960 and 1961–2024). The regression is estimated separately for each period, and we compare the slope coefficients.

The hypothesis is formulated as:

$$H_0: \beta_{\text{early}} = \beta_{\text{late}} \text{against} H_1: \beta_{\text{late}} > \beta_{\text{early}}.$$

The null hypothesis implies that the rate of temperature increase remained constant over time, while the alternative suggests that the warming trend has intensified in the later period.

For each subperiod, a linear regression of temperature on time is estimated. The slope coefficient βmeasures the average yearly change in temperature. The comparison of slopes across periods allows us to assess whether the trend has strengthened.

We discuss results for one city: Maastricht. However, the results for De Bilt and Eelde follow the same pattern. This consistency across cities suggests that the observed increase in the temperature trend is not location-specific but reflects a broader regional change.

Regression Results — Maastricht (Annual Data, Split Sample)

Period	Slope (β, °C/year)	SE(β)	t-statistic	p-value	95% CI for β
1907–1960	0.014074	0.004917	2.8620	0.0061	[0.004206 , 0.023941]
1961–2024	0.041081	0.004210	9.7568	3.82e-14	[0.032664 , 0.049498]

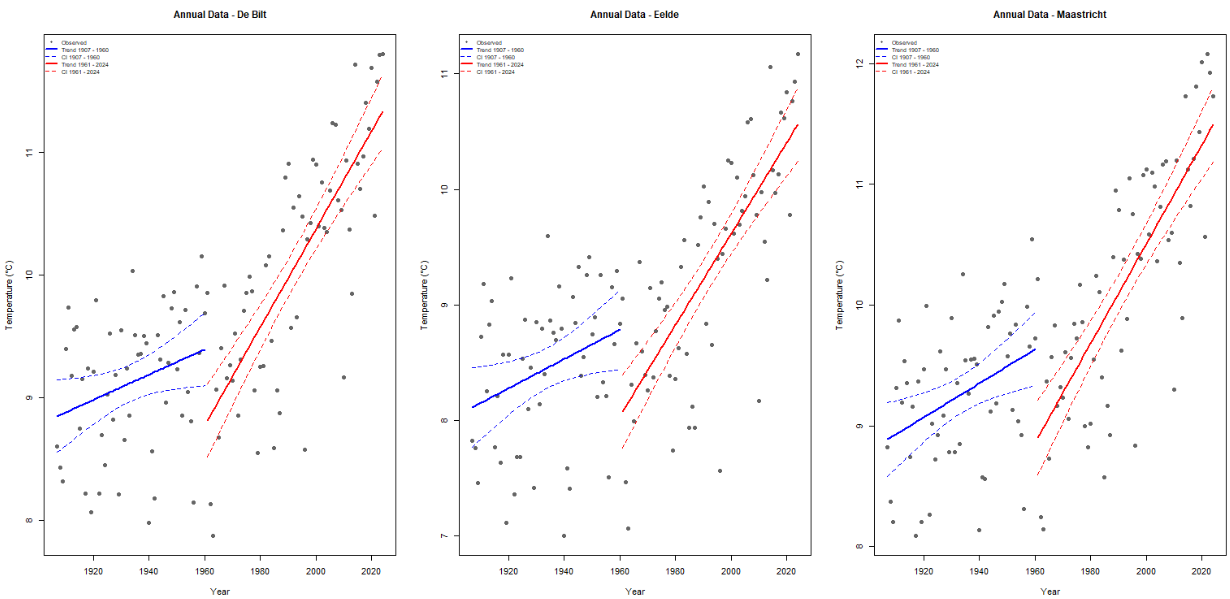

Figure 7: Regression graphs for split samples annual data

The results show a clear increase in the magnitude of the temperature trend between the two periods. In the early period, the estimated slope is relatively moderate (approximately 0.014°C per year), indicating a gradual increase in temperature. In contrast, the slope in the late period rises to approximately 0.041°C per year, which is nearly three times larger.

Both estimates are statistically significant, as indicated by the p-values and confidence intervals that lie above zero. However, the much larger t-statistic in the later period reflects a stronger and more precisely estimated trend.

This difference is also clearly visible in Figure 7. The regression line for the late period (red) is noticeably steeper than that of the early period (blue), illustrating the acceleration in temperature increases over time.

Overall, the results provide strong evidence against the null hypothesis and support the conclusion that the warming trend has intensified after 1960.

Monthly extension

Regression Results — Maastricht (Smoothed Monthly Data, Split Sample)

Period	Slope (β, °C/month)	SE(β)	t-statistic	p-value	95% CI for β
1907–1960	0.001156	0.000127	9.0935	< 2.22e-16	[0.000907 , 0.001406]
1961–2024	0.003448	0.000102	33.7997	< 2.22e-16	[0.003248 , 0.003648]

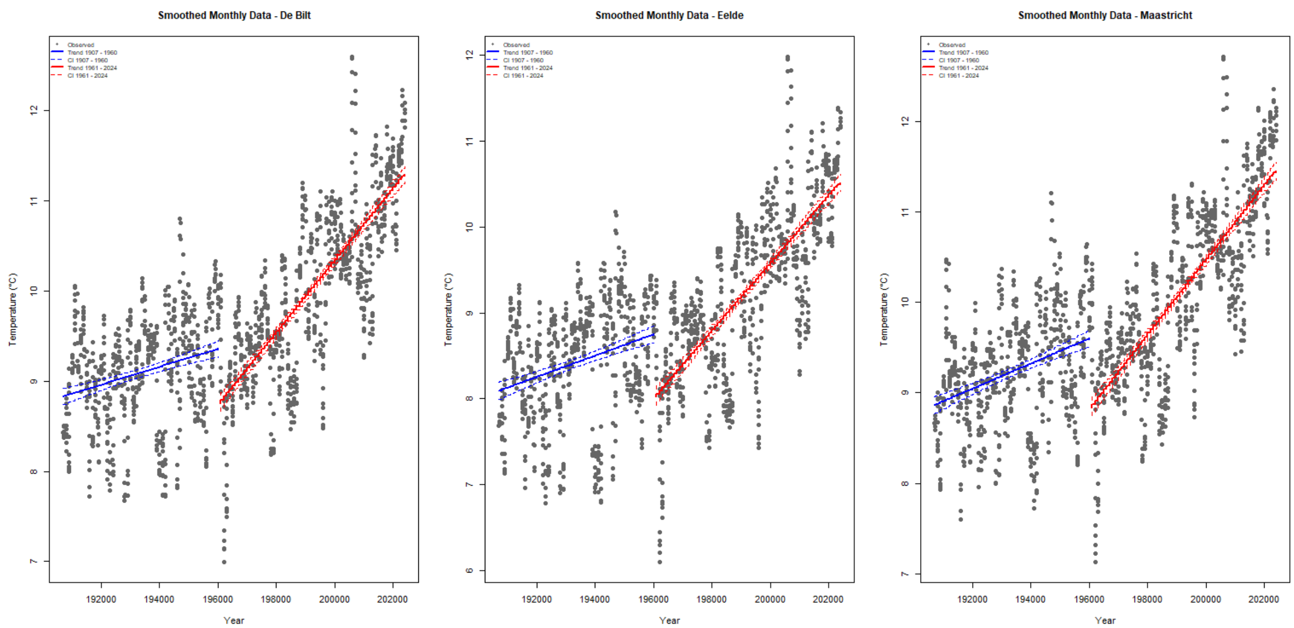

Figure 8: Regression graphs for split samples smoothed monthly data

The results for the smoothed monthly data confirm the same pattern observed in the annual analysis, but with substantially higher accuracy. In both periods, the estimated slopes are positive and highly statistically significant.

The magnitude of the trend increases noticeably between periods. In the early period, the slope is approximately 0.0012°C per month, while in the late period it rises to about 0.0034°C per month. This indicates a clear intensification of the warming trend after 1960, consistent with the findings from the annual data.

The smaller numerical values of β compared to the annual results are due to the different time scale. When converted to yearly terms (multiplying by 12), the estimates are directly comparable to those from the annual data and confirm the same magnitude of warming.

In addition, the t-statistics are substantially larger and the confidence intervals narrower than in the annual case. This reflects the much larger number of observations in the monthly dataset, which leads to more precise estimates of the trend.

Bootstrap Analysis

The paired bootstrap method is used to assess the robustness and statistical significance of the estimated temperature trends. This method is based on the standard nonparametric bootstrap, in which observations are repeatedly sampled with replacement from the original dataset.

In this setting, the data consist of pairs (X_i, Y_i), where X_i denotes time and Y_i the corresponding temperature. Instead of resampling each variable individually, the bootstrap procedure selects entire pairs. This ensures that the relationship between time and temperature is preserved within each resampled dataset.

The validity of this method formally relies on the assumption that the observed pairs are independent and identically distributed. However, as shown in the residual analysis, temperature data exhibit clear temporal dependence, particularly in the monthly series, where strong autocorrelation is present. This implies that the independence assumption is not strictly satisfied.

Despite this limitation, the bootstrap remains a useful tool in this context. Unlike standard OLS inference, which is highly sensitive to violations of independence, the bootstrap provides a more flexible, data-driven approximation of the sampling distribution.

Hypothesis Testing Framework

The bootstrap procedure is used to test for the presence of a positive trend in temperature over time. The hypotheses are formulated as:

$$H_0: \beta \leq 0 \text{against} H_1: \beta > 0,$$

where βrepresents the slope coefficient from the linear regression of temperature on time.

To assess the strength of the estimated trend, the test statistic is defined as:

$$Q = \frac{\hat{\beta}}{\widehat{se}(\hat{\beta})}.$$

A large value of Qindicates that the estimated slope is large relative to its sampling variability.

To approximate the sampling distribution of the test statistic, a large number of bootstrap samples is generated. For each resampled dataset, the regression model is re-estimated, resulting in bootstrap estimates $\hat{\beta}^*$. Based on these, a centered bootstrap statistic is constructed:

$$Q^* = \frac{\hat{\beta}^* - \hat{\beta}}{\widehat{se}(\hat{\beta}^*)}.$$

The critical value for the test is obtained as the appropriate quantile of this distribution, and the p-value is computed as the proportion of bootstrap statistics exceeding the observed value of Q.

The null hypothesis is rejected if the observed test statistic exceeds the bootstrap critical value, or equivalently if the bootstrap p-value is below the chosen significance level.

Rejection of the null hypothesis implies that the estimated slope is significantly positive, indicating the presence of an upward temperature trend that is unlikely to be explained by random variation alone.

Results

To determine whether a structural change has occurred in temperature trends, the dataset is divided into two periods: the early period (1907–1960) and the late period (1961–2024), just as in the regression approach. A paired bootstrap method is applied to each subsample to estimate the slope coefficient and test for the positive trend.

The results are similar across all cities, so we are focusing on implementing the data for one city: Maastricht.

Annua data results by samples for Maastricht:

Period	β (Slope)	β* (Mean)	Q statistic	Critical Value	P - value	95% CI for β
1907-1960	0.014074	0.014145	2.8620	1.7289	0.0055	[0.004129 , 0.023860]
1961-2024	0.041081	0.041084	9.7568	1.6039	0.0000	[0.032649 , 0.048760]

In the early period, the estimated slope is $\beta = 0.0141$, suggesting a modest upward trend in temperature. The corresponding bootstrap estimate $\beta^*$is very close to this value, confirming the stability of the estimator. The bootstrap confidence interval does not include zero, indicating that the trend is statistically significant. This is further supported by the test statistic $Q = 2.862$, which exceeds the bootstrap critical value of 1.729. The associated p-value (0.0055) provides strong evidence against the null hypothesis of no positive trend.

In the late period, the estimated slope increases substantially to $\beta = 0.0411$, indicating a much stronger warming trend. Again, the bootstrap estimate closely matches the original estimate, and the confidence interval lies entirely above zero. The test statistic is significantly larger $Q = 9.757$ and far exceeds the critical value 1.604, with a p-value effectively equal to zero. This provides very strong evidence of a positive trend in the later period.

Overall, the results suggest that while a statistically significant warming trend is already present in the early period, the magnitude of this trend increases considerably after 1960. This supports the presence of a structural change in temperature dynamics, with more intense warming in the second half of the sample.

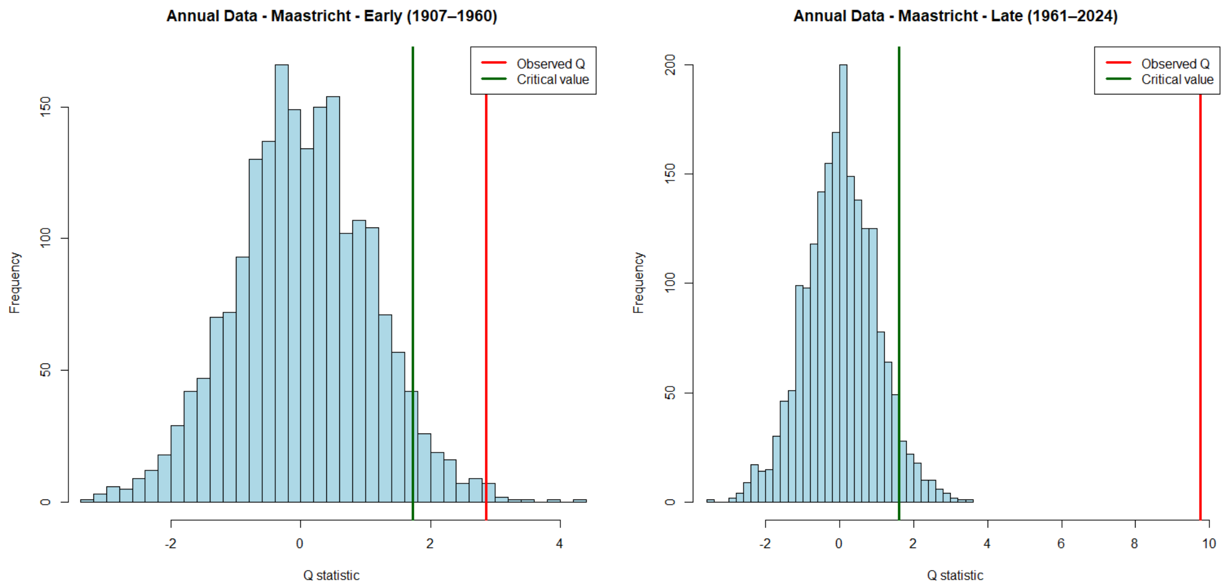

Figure 9: Bootstrap annual data graphs for Maastricht

The graphical representation of the bootstrap distributions further supports these findings (see Figure 9). In both periods, the histogram illustrates the empirical distribution of the centered bootstrap statistic Q^*, while the vertical lines indicate the observed test statistic and the corresponding critical value.

As shown, the observed Q-statistic in the late period lies far in the right tail of the bootstrap distribution and is well above the critical value, indicating very strong evidence against the null hypothesis. In contrast, in the early period the statistic is only moderately above the critical threshold, suggesting a weaker, though still statistically significant, positive trend.

Monthly data extension

Smoothed Monthly data results by samples for Maastricht

Period	β (Slope)	β* (Mean)	Q statistic	Critical Value	P - value	95% CI for β
1907-1960	0.001156	0.001155	9.0935	1.6639	0.0000	[0.000896 , 0.001405]
1961-2024	0.003448	0.003449	33.7997	1.5778	0.0000	[0.003249 , 0.003649]

The results for the smoothed monthly data confirm the same pattern as observed for the annual data, but with much stronger statistical significance. In both periods, the estimated trends are positive and highly significant, with p-values effectively equal to zero and confidence intervals well above zero. Compared to the annual data, the Q-statistics are substantially larger, reflecting the increased accuracy from the higher-frequency data.

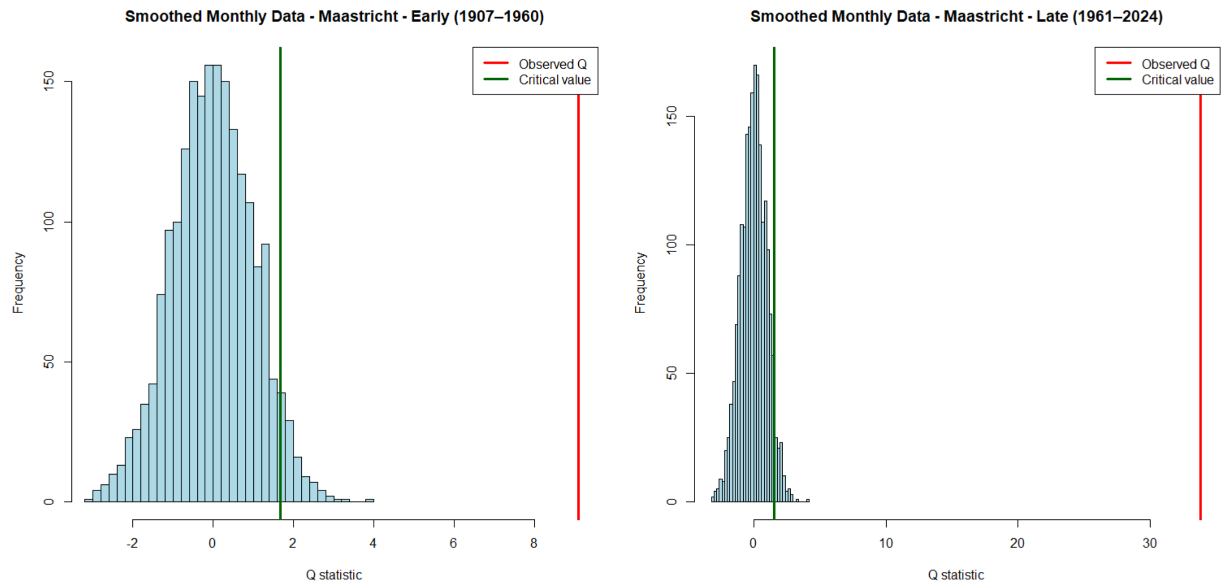

Figure 10: Bootstrap smoothed monthly data graphs for Maastricht

As shown in Figure 10, the observed Q-statistic in the late period is again far in the right tail of the bootstrap distribution and even more extreme than in the early period, indicating a very strong and robust upward trend after 1960.

Bootstrap Conclusion

The bootstrap analysis provides strong and consistent evidence against the null hypothesis $H_0: \beta \leq 0$ for all considered cases. In both the annual and smoothed monthly datasets, and across both time periods, the observed test statistics exceed the corresponding bootstrap critical values, while the p-values are either very small or effectively zero.

This implies that the null hypothesis of no positive trend is rejected in all cases, providing robust statistical evidence of a positive temperature trend.

Importantly, this result is consistent across all cities (Maastricht, De Bilt, and Eelde), indicating that the observed warming pattern is not location-specific.

Furthermore, the magnitude of the estimated trend increases substantially after 1960. This effect is particularly pronounced in the smoothed monthly data, where higher-frequency observations lead to more precise estimates and stronger statistical significance.

Overall, the bootstrap results confirm the existence of strengthening upward temperature trend over time across all considered regions.

Conclusion

This paper investigated whether the temperate in three Dutch cities, Maastricht, De Bilt, and Eelde, have risen in an impactful way between 1907 and 2024, and whether this trend intensified from 1960 onwards. We can now draw some conclusions.

Firstly, looking at both rolling and block averages, temperatures appeared fairly stable during the first half of the sample, roughly up to the 1970s. After that, a clear warming acceleration becomes visible across all three cities, most noticeably from the 1980s onward. Maastricht consistently sat at the top in terms of temperature levels, while Eelde remained the coolest throughout.

Moving to the regression analysis, running OLS over the entire period gave us the following results. Slope estimates were around 0.019°C and p-values effectively at zero. The confidence intervals were all above zero, suggesting strong evidence of an upward trend in temperature. Alongside this we also ran the regression for the monthly data set and found similar results. Since the results were consistent across Maastricht, De Bilt and Eelde, it is unlikely that local factors alone can explain the warming, pointing instead to a broader regional trend.

To further investigate this, we split the data at 1960 and estimated the regression for each of the two subperiods. For Maastricht, the slope went from roughly 0.014°C per year in the early period to 0.041°C per year after 1960. The same pattern was observed for the other cities. The bootstrap analysis supports this. In both subperiods, the observed test statistics exceeded the bootstrap critical values and the p-values were near zero. Hence the null hypothesis of no positive trend is rejected in every case.

Putting this altogether, our data points in the same direction. Temperatures in the Netherlands have been steadily rising over the past century and noticeably rose further from 1960 onwards, which lines up with the period of rapid growth in car ownership and commercial aviation.

References

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APPENDIX

Appendix 1: Assumptions outputs for De Bilt and Eelde

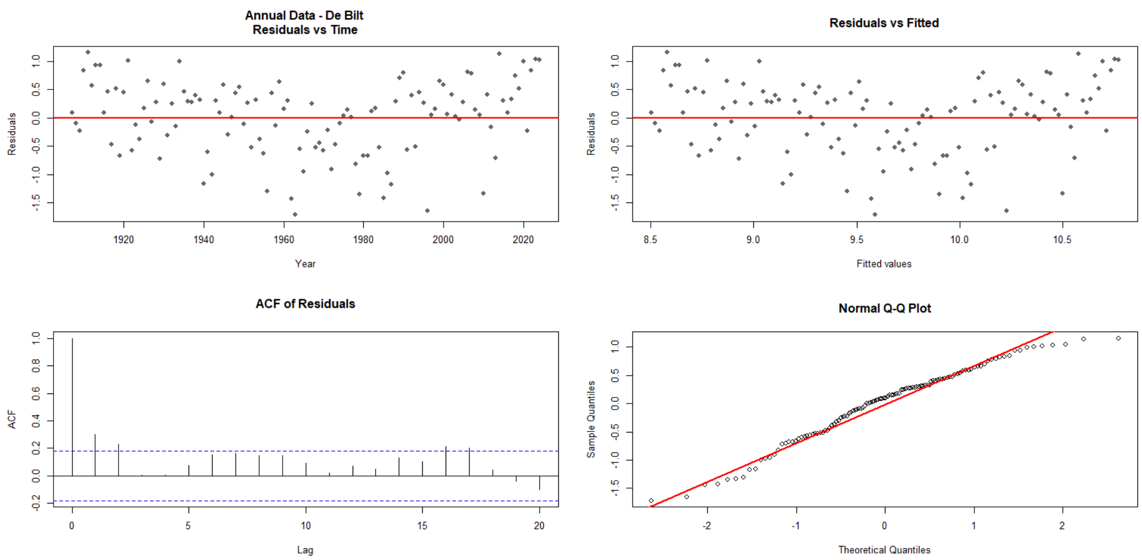
ASSUMPTION CHECK: Annual Data

Location: De Bilt

Lag-1 autocorrelation: 0.3001

ACF bound (±2/sqrt(n)): ±0.1841

QQ correlation: 0.9860

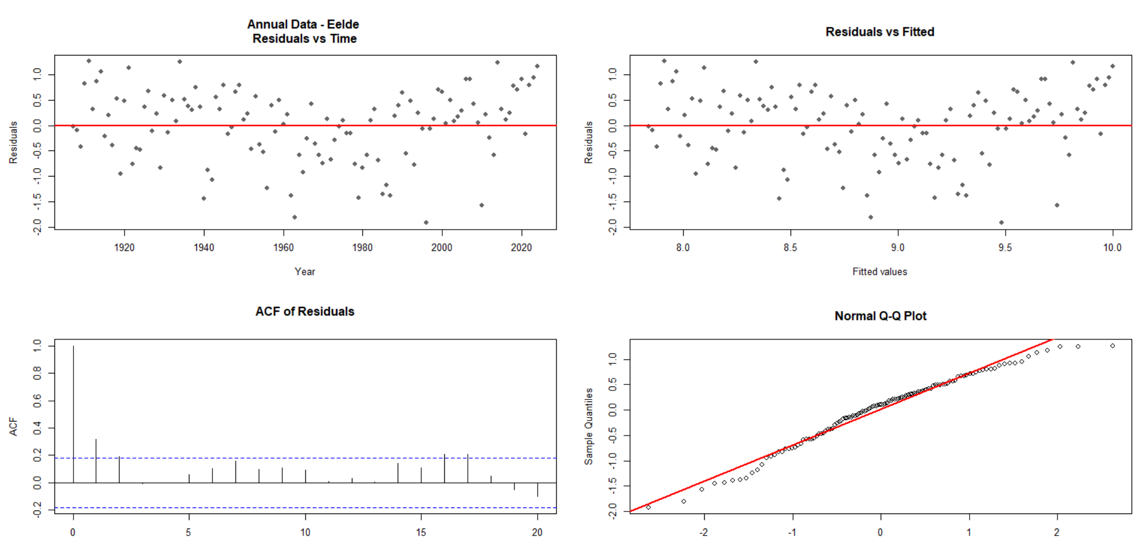


Location: Eelde

Lag-1 autocorrelation: 0.3173

ACF bound (±2/sqrt(n)): ±0.1841

QQ correlation: 0.9870



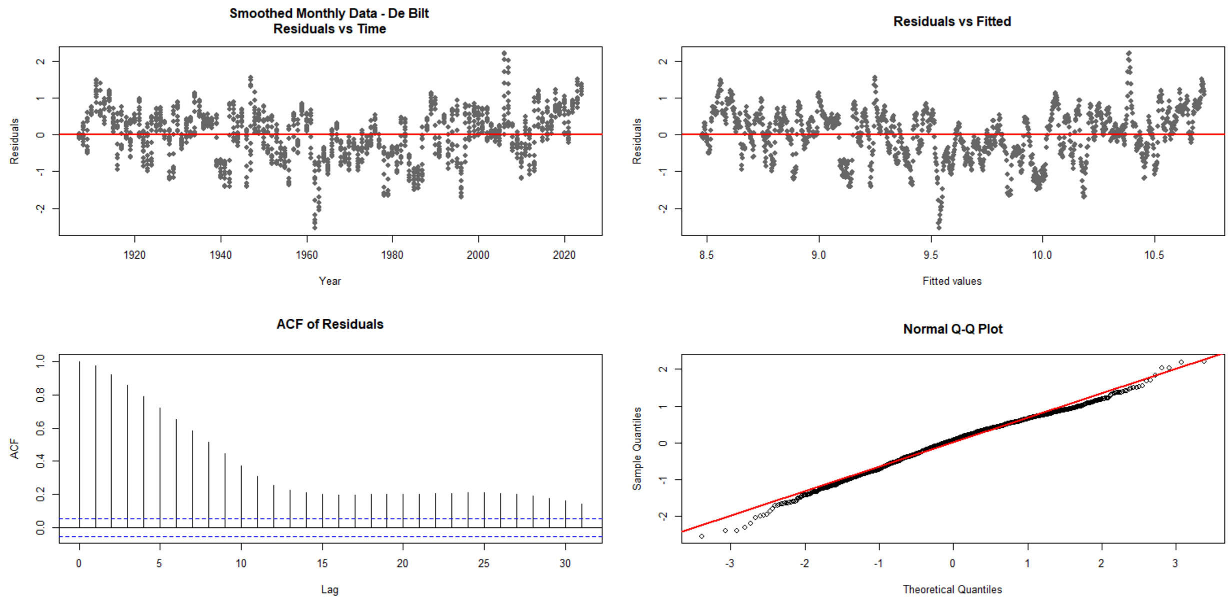
ASSUMPTION CHECK: Smoothed Monthly Data

Location: De Bilt

Lag-1 autocorrelation: 0.9744

ACF bound (±2/sqrt(n)): ±0.0534

QQ correlation: 0.9946

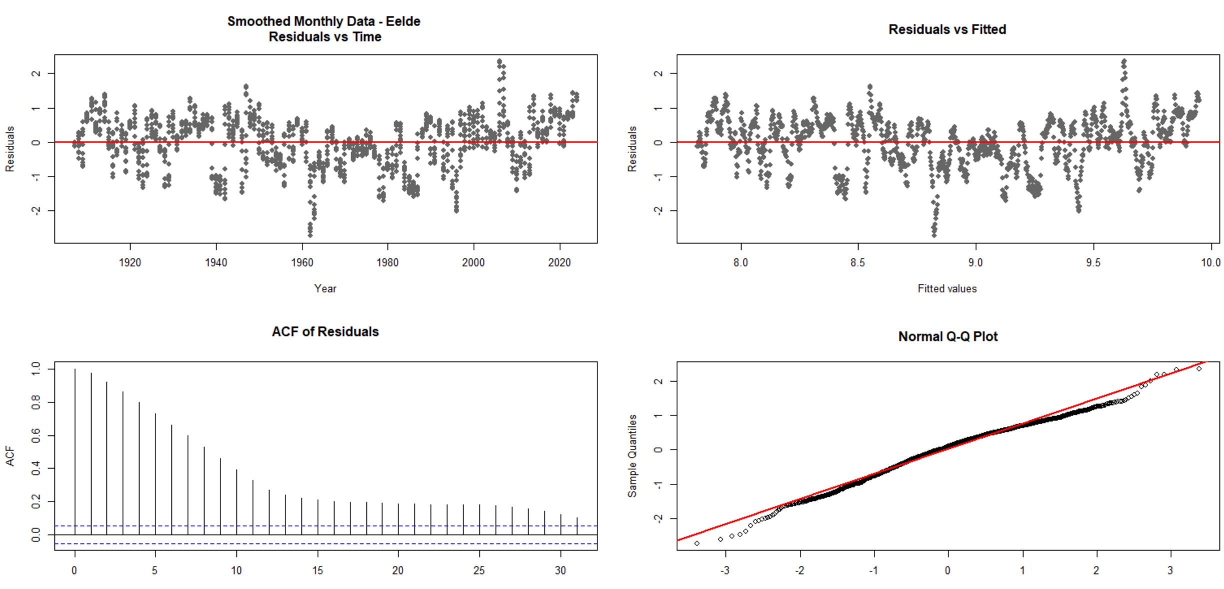


Location: Eelde

Lag-1 autocorrelation: 0.9760

ACF bound (±2/sqrt(n)): ±0.0534

QQ correlation: 0.9919



Appendix 2: Bootstrap outputs for De Bilt and Eelde

BOOTSTRAP: Annual Data

City: De.Bilt

EARLY PERIOD

beta = 0.010264 | beta_star = 0.010320

Q = 2.1374 | crit = 1.6528 | p = 0.022000

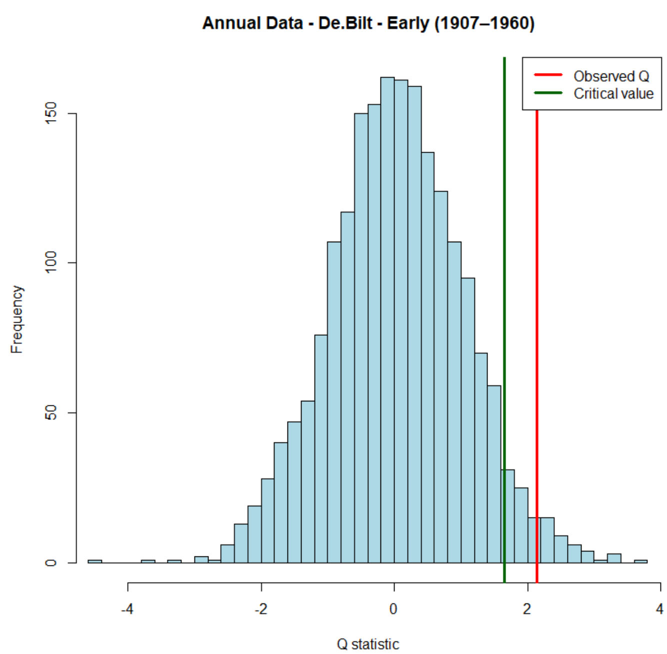
CI_beta = [0.000749 , 0.019289]

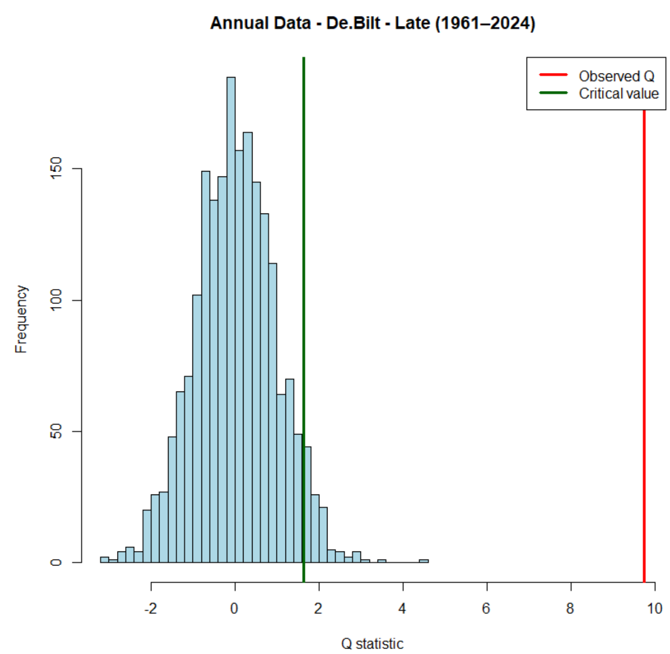
LATE PERIOD

beta = 0.039953 | beta_star = 0.039940

Q = 9.7526 | crit = 1.6310 | p = 0.000000

CI_beta = [0.032023 , 0.047017]





City: Eelde

EARLY PERIOD

beta = 0.012646 | beta_star = 0.012667

Q = 2.2764 | crit = 1.6358 | p = 0.009000

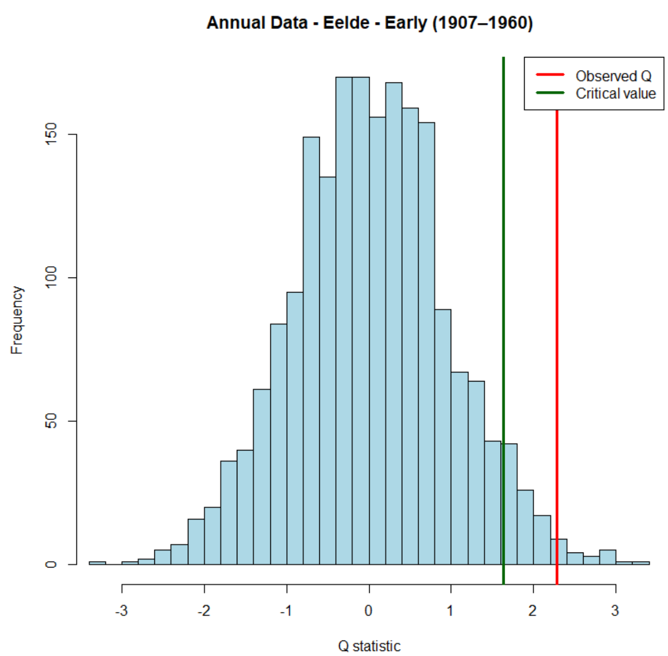
CI_beta = [0.002272 , 0.022677]

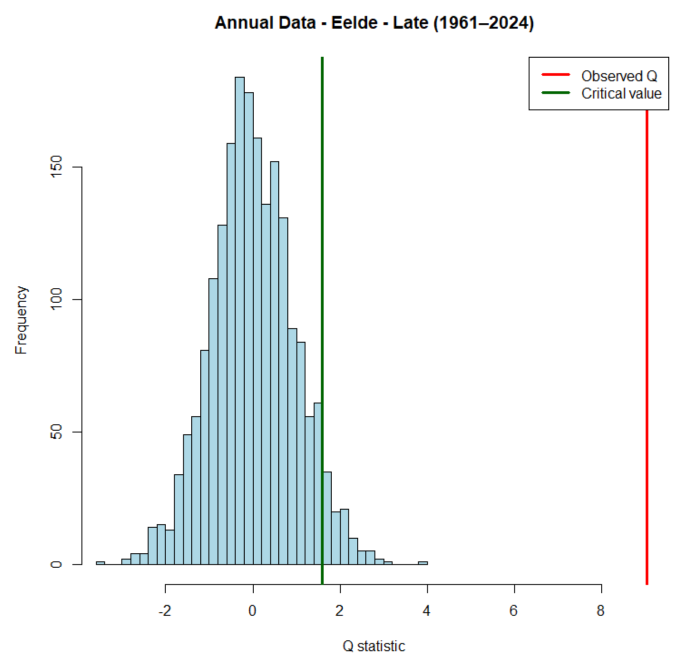
LATE PERIOD

beta = 0.039406 | beta_star = 0.039307

Q = 9.0478 | crit = 1.5995 | p = 0.000000

CI_beta = [0.030824 , 0.047153]





BOOTSTRAP: Smoothed Monthly Data

City: De.Bilt

EARLY PERIOD

beta = 0.000829 | beta_star = 0.000826

Q = 6.7215 | crit = 1.5887 | p = 0.000000

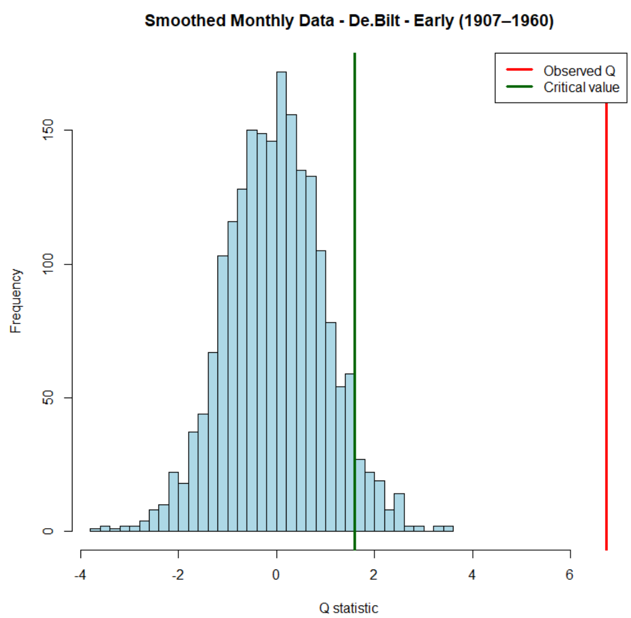
CI_beta = [0.000579 , 0.001072]

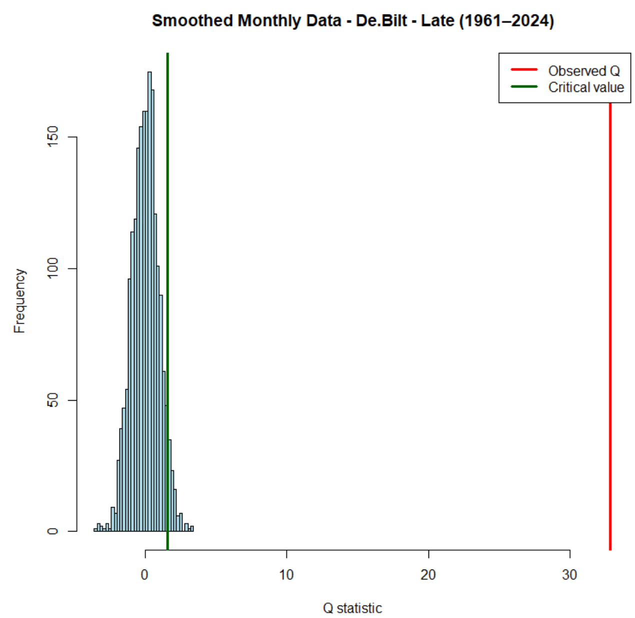
LATE PERIOD

beta = 0.003335 | beta_star = 0.003336

Q = 32.8636 | crit = 1.5542 | p = 0.000000

CI_beta = [0.003154 , 0.003523]





City: Eelde

EARLY PERIOD

beta = 0.001034 | beta_star = 0.001034

Q = 7.4114 | crit = 1.5250 | p = 0.000000

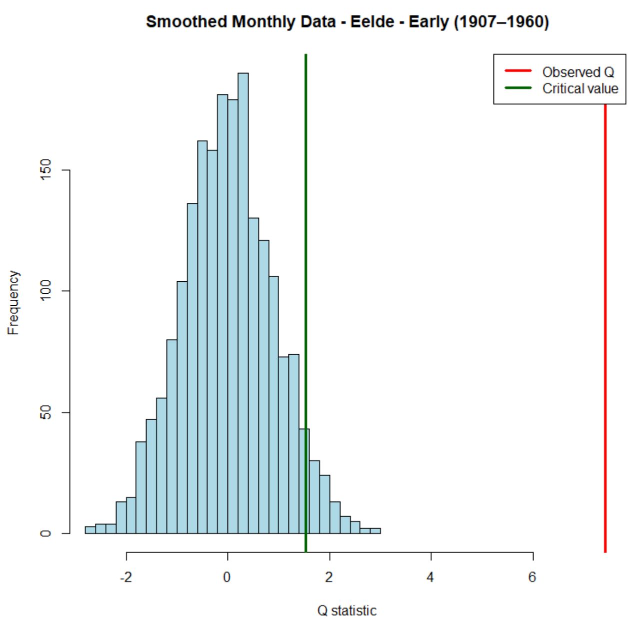
CI_beta = [0.000791 , 0.001281]

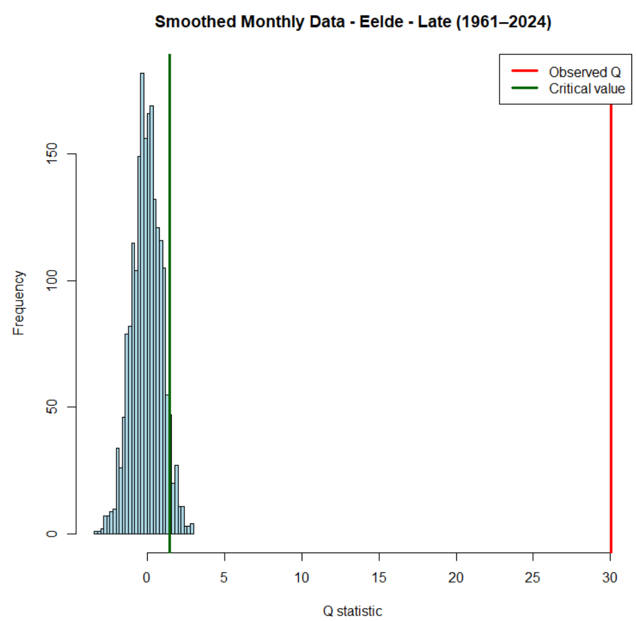
LATE PERIOD

beta = 0.003280 | beta_star = 0.003278

Q = 30.0421 | crit = 1.4850 | p = 0.000000

CI_beta = [0.003076 , 0.003484]





Appendix 3: Averages Code

```
rm(list=ls())


#rolling and block averages function

run_averages_analysis <- function(data, dataset_name, year_col = "Date") {


 year <- data[[year_col]]


 #handle different date formats

 if (max(year) > 10000) {

  #monthly data format (YYYYMM)

  year_plot <- year / 100

  window_size <- 120  # 10 years * 12 months

  block_size <- 120   # 10 years * 12 months

 } else {

  #annual data format (YYYY)

  year_plot <- year

  window_size <- 10   # 10 years

  block_size <- 10    # 10 years

 }


 bilt <- data$De.Bilt

 eelde <- data$Eelde

 maas <- data$Maastricht


 n <- length(year)
```

```
cat("\n\n================================\n")
cat("Dataset:", dataset_name, "\n")
cat("Observations:", n, "\n")
cat("Rolling window:", window_size, "observations\n")
cat("Block size:", block_size, "observations\n")
cat("================================\n")

#rolling averages
roll_bilt <- rep(NA, n)
roll_eelde <- rep(NA, n)
roll_maas <- rep(NA, n)

for(i in window_size:n){
  roll_bilt[i] <- mean(bilt[(i-window_size+1):i])
  roll_eelde[i] <- mean(eelde[(i-window_size+1):i])
  roll_maas[i] <- mean(maas[(i-window_size+1):i])
}

#plotting rolling averages
plot(year_plot, bilt,
     type="l",
     col="grey70",
     ylim=range(c(bilt, eelde, maas)),
     xlab="Year",
     ylab="Temperature (°C)",
     main=paste(dataset_name, "- Rolling Averages"))
```

```
lines(year_plot, eelde, col="grey50")

lines(year_plot, maas, col="grey30")


lines(year_plot, roll_bilt, col="blue", lwd=2)

lines(year_plot, roll_eelde, col="red", lwd=2)

lines(year_plot, roll_maas, col="darkgreen", lwd=2)


legend("topleft",

    legend=c("De Bilt (raw)", "Eelde (raw)", "Maastricht (raw)",

        paste("De Bilt", window_size/12, "y avg"),

        paste("Eelde", window_size/12, "y avg"),

        paste("Maastricht", window_size/12, "y avg")),

    col=c("grey70","grey50","grey30","blue","red","darkgreen"),

    lty=1,

    lwd=c(1,1,1,2,2,2),

    cex=0.8)


#block averages

num_blocks <- floor(n / block_size)


block_year <- rep(NA, num_blocks)

block_bilt <- rep(NA, num_blocks)

block_eelde <- rep(NA, num_blocks)

block_maas <- rep(NA, num_blocks)


for(i in 1:num_blocks){

  start <- (i-1)*block_size + 1
```

```
  end <- i*block_size


  block_year[i] <- mean(year_plot[start:end])
  block_bilt[i] <- mean(bilt[start:end])
  block_eelde[i] <- mean(eelde[start:end])
  block_maas[i] <- mean(maas[start:end])
}


cat("\nNumber of blocks:", num_blocks, "\n")


#plotting block averages
plot(block_year, block_bilt,
    type="b",
    pch=16,
    col="blue",
    ylim=range(c(block_bilt, block_eelde, block_maas)),
    xlab="Year",
    ylab="Average Temperature (°C)",
    main=paste(dataset_name, "- Block Averages"))


lines(block_year, block_eelde, type="b", pch=16, col="red")
lines(block_year, block_maas, type="b", pch=16, col="darkgreen")


legend("topleft",
      legend=c("De Bilt", "Eelde", "Maastricht"),
      col=c("blue", "red", "darkgreen"),
      lty=1,
```

```
        pch=16)
    }



    #load annual data
    data_annual <- read.csv("AnnualTemp.csv", sep=";", dec=",")
    colnames(data_annual) <- c("Date", "De.Bilt", "Eelde", "Maastricht")

    #load monthly data
    data_monthly <- read.csv("SmoothedMonthlyTemp.csv", sep=";", dec=",")
    colnames(data_monthly) <- c("Date", "De.Bilt", "Eelde", "Maastricht")

    #run averages analysis
    par(mfrow = c(1, 2))
    run_averages_analysis(data_annual, "Annual Data")

    par(mfrow = c(1, 2))
    run_averages_analysis(data_monthly, "Smoothed Monthly Data")

    par(mfrow = c(1, 1))
```

Appendix 4: Assumptions test code

```
rm(list=ls())


# OLS WITH RESIDUALS

ols_with_residuals <- function(x, y) {


  valid <- complete.cases(x, y)

  x <- x[valid]

  y <- y[valid]


  n <- length(y)


  X <- cbind(1, x)

  beta_hat <- solve(t(X) %*% X) %*% t(X) %*% y

  y_hat   <- X %*% beta_hat

  e       <- y - y_hat


  list(beta = beta_hat,

       yhat = y_hat,

       e    = e,

       n    = n)

}


# MAIN ASSUMPTION CHECK FUNCTION

run_assumption_checks <- function(data, dataset_name,

                                  year_col = "Date") {
```

```
# check columns
required_cols <- c(year_col, "De.Bilt", "Eelde", "Maastricht")
missing_cols <- setdiff(required_cols, colnames(data))

if (length(missing_cols) > 0) {
  cat("\nERROR:", dataset_name, "\n")
  cat("Missing columns:", paste(missing_cols, collapse=", "), "\n")
  return(invisible(NULL))
}

year <- data[[year_col]]

# handle monthly vs annual
if (max(year) > 10000) {
  year_numeric <- floor(year / 100)
} else {
  year_numeric <- year
}

locations <- list(
  "De Bilt"    = data$De.Bilt,
  "Eelde"      = data$Eelde,
  "Maastricht" = data$Maastricht
)

cat("\n\n==================================\n")
cat("ASSUMPTION CHECK:", dataset_name, "\n")
```

```
cat("================================\n")

for (loc_name in names(locations)) {

  y <- locations[[loc_name]]
  x <- seq_along(y)

  model <- ols_with_residuals(x, y)

  if (is.null(model)) next

  e <- model$e
  y_hat <- model$yhat
  n <- model$n

  cat("\n--------------------------------\n")
  cat("Location:", loc_name, "\n")
  cat("--------------------------------\n")

  # PRINT NUMERIC CHECKS

  # lag-1 autocorrelation
  acf_vals <- acf(e, plot = FALSE)
  acf_lag1 <- acf_vals$acf[2]
  bound <- 2 / sqrt(n)

  # QQ correlation
```

```
qq_corr <- cor(sort(e), qnorm(ppoints(length(e))))

cat(sprintf("Lag-1 autocorrelation: %.4f\n", acf_lag1))
cat(sprintf("ACF bound (±2/sqrt(n)): ±%.4f\n", bound))
cat(sprintf("QQ correlation: %.4f\n", qq_corr))

# PLOTS

par(mfrow = c(2, 2))

# Residuals vs Time (Linearity)
plot(year_numeric, e,
    pch = 16, col = "grey40",
    xlab = "Year", ylab = "Residuals",
    main = paste(dataset_name, "-", loc_name, "\nResiduals vs Time"))
abline(h = 0, col = "red", lwd = 2)

#  Residuals vs Fitted (Homoskedasticity)
plot(y_hat, e,
    pch = 16, col = "grey40",
    xlab = "Fitted values", ylab = "Residuals",
    main = "Residuals vs Fitted")
abline(h = 0, col = "red", lwd = 2)

#  ACF (Serial Independence)
acf(e,
   main = "ACF of Residuals")
```

```r
    #  QQ plot (Normality)
    qqnorm(e,
           main = "Normal Q-Q Plot")
    qqline(e, col = "red", lwd = 2)

    par(mfrow = c(1,1))
  }
}

# LOAD DATA
data_annual <- read.csv("AnnualTemp.csv", sep=";", dec=",")
colnames(data_annual) <- c("Date", "De.Bilt", "Eelde", "Maastricht")

data_monthly <- read.csv("SmoothedMonthlyTemp.csv", sep=";", dec=",")
colnames(data_monthly) <- c("Date", "De.Bilt", "Eelde", "Maastricht")

# RUN
run_assumption_checks(data_annual,  "Annual Data")
run_assumption_checks(data_monthly, "Smoothed Monthly Data")
```

Appendix 5: Regression full data code

```
rm(list=ls())


#regression function

run_whole_sample_regression <- function(data, dataset_name, year_col = "Date") {


  year <- data[[year_col]]


  #handle different date formats

  if (max(year) > 10000) {

    #monthly data format (YYYYMM): convert to decimal years for plotting

    year_plot <- year / 100

  } else {

    #annual data format (YYYY)

    year_plot <- year

  }


  locations <- list("De Bilt" = data$De.Bilt,

                    "Eelde" = data$Eelde,

                    "Maastricht" = data$Maastricht)


  n <- nrow(data)

  x <- 1:n

  X <- cbind(1, x)


  cat("\n\n=====================================\n")

  cat("Dataset:", dataset_name, "\n")
```

```
cat("Observations:", n, "\n")
cat("===================================\n")

par(mfrow = c(1, 3))

for (loc_name in names(locations)) {
  y <- locations[[loc_name]]

  beta_hat <- solve(t(X) %*% X) %*% t(X) %*% y
  y_hat    <- X %*% beta_hat
  e        <- y - y_hat
  sigma2   <- sum(e^2) / (n - 2)
  var_beta <- sigma2 * solve(t(X) %*% X)
  se_beta  <- sqrt(var_beta[2,2])
  t_stat   <- beta_hat[2] / se_beta
  p_value  <- 2 * (1 - pt(abs(t_stat), df = n - 2))

  Sxx    <- sum((x - mean(x))^2)
  t_crit <- qt(0.975, df = n - 2)
  se_fit <- sqrt(sigma2 * (1/n + (x - mean(x))^2 / Sxx))
  upper  <- y_hat + t_crit * se_fit
  lower  <- y_hat - t_crit * se_fit

  cat("\n--------------------------------\n")
  cat("Location:", loc_name, "\n")
  cat(sprintf("Intercept:   %.4f\n", beta_hat[1]))
  cat(sprintf("Slope:       %.4f\n", beta_hat[2]))
```

```
    cat(sprintf("SE:         %.4f\n", se_beta))
    cat(sprintf("t-statistic: %.4f\n", t_stat))
    cat(sprintf("p-value:     %s\n",   format.pval(p_value)))
    cat(sprintf("95%% CI:     [%.4f, %.4f]\n",
              beta_hat[2] - t_crit * se_beta,
              beta_hat[2] + t_crit * se_beta))


    plot(year_plot, y, pch = 16, col = "grey40",
         xlab = "Year", ylab = "Temperature (°C)",
         main = paste(dataset_name, "-", loc_name))
    lines(year_plot, y_hat, col = "red",  lwd = 2)
    lines(year_plot, upper, col = "blue", lty = 2)
    lines(year_plot, lower, col = "blue", lty = 2)
    legend("topleft",
           legend = c("Observed", "Trend", "95% CI"),
           col = c("grey40", "red", "blue"),
           pch = c(16, NA, NA), lty = c(NA, 1, 2),
           lwd = c(NA, 2, 1), bty = "n")
  }


  par(mfrow = c(1, 1))
}



#load annual data
data_annual <- read.csv("AnnualTemp.csv", sep=";", dec=",")
colnames(data_annual) <- c("Date", "De.Bilt", "Eelde", "Maastricht")
```

```
#load monthly data

data_monthly <- read.csv("SmoothedMonthlyTemp.csv", sep=";", dec=",")

colnames(data_monthly) <- c("Date", "De.Bilt", "Eelde", "Maastricht")


#run regression

run_whole_sample_regression(data_annual, "Annual Data")

run_whole_sample_regression(data_monthly, "Smoothed Monthly Data")
```

Appendix 6: Regression by samples code

```
rm(list=ls())

ols_estimate <- function(x, y) {

 # remove missing values
 valid <- complete.cases(x, y)
 x <- x[valid]
 y <- y[valid]
 n <- length(y)

 # Check if we have enough data points
 if (n < 3) {
  cat("ERROR: Not enough valid data points for regression\n")
  return(NULL)
 }

 X       <- cbind(1, x)
 beta_hat <- solve(t(X) %*% X) %*% t(X) %*% y
 y_hat    <- X %*% beta_hat
 e        <- y - y_hat
 sigma2   <- sum(e^2) / (n - 2)
 var_beta <- sigma2 * solve(t(X) %*% X)
 se_beta  <- sqrt(var_beta[2,2])
 t_stat   <- beta_hat[2] / se_beta
 p_value  <- 2 * (1 - pt(abs(t_stat), df = n - 2))
 t_crit   <- qt(0.975, df = n - 2)
```

```
  Sxx    <- sum((x - mean(x))^2)
  se_fit  <- sqrt(sigma2 * (1/n + (x - mean(x))^2 / Sxx))

  list(beta  = beta_hat,
     se    = se_beta,
     t     = t_stat,
     p     = p_value,
     ci_beta= c(beta_hat[2] - t_crit * se_beta,
          beta_hat[2] + t_crit * se_beta),
     yhat  = y_hat,
     upper  = y_hat + t_crit * se_fit,
     lower  = y_hat - t_crit * se_fit)
}

# GENERAL ANALYSIS FUNCTION
run_temperature_analysis <- function(data, dataset_name,
                   year_col = "Date",
                   split_year = 1960,
                   start_year = 1907,
                   end_year = 2024) {

  # Check if required columns exist
  required_cols <- c(year_col, "De.Bilt", "Eelde", "Maastricht")
  missing_cols <- setdiff(required_cols, colnames(data))

  if (length(missing_cols) > 0) {
```

```
    cat("\nERROR:", dataset_name, "\n")
    cat("Missing columns:", paste(missing_cols, collapse=", "), "\n")
    cat("Available columns:", paste(colnames(data), collapse=", "), "\n")
    return(invisible(NULL))
  }

  year <- data[[year_col]]

  # For annual data: years are already correct
  # For monthly data: dates are YYYYMM format, so extract year
  if (max(year) > 10000) {
    # Monthly data format (YYYYMM)
    year_numeric <- floor(year / 100)
  } else {
    # Annual data format (YYYY)
    year_numeric <- year
  }

  locations <- list(
    "De Bilt"    = data$De.Bilt,
    "Eelde"      = data$Eelde,
    "Maastricht"  = data$Maastricht
  )

  early_idx <- year_numeric <= split_year
  late_idx  <- year_numeric > split_year
```

```
early_label <- paste(start_year, "-", split_year)
late_label  <- paste(split_year + 1, "-", end_year)

cat("\n\n==================================\n")
cat("Dataset:", dataset_name, "\n")
cat("Early period:", early_label, "\n")
cat("Late period :", late_label, "\n")
cat("Total observations:", nrow(data), "\n")
cat("==================================\n")

par(mfrow = c(1,3))

for (loc_name in names(locations)) {

  y <- locations[[loc_name]]

  m1 <- ols_estimate(seq_along(y[early_idx]), y[early_idx])
  m2 <- ols_estimate(seq_along(y[late_idx]),  y[late_idx])

  # Check if regression was successful
  if (is.null(m1) || is.null(m2)) {
    cat("\nSkipping", loc_name, "due to insufficient data\n")
    next
  }

  cat("\n--------------------------------\n")
  cat(loc_name, "|", early_label, "\n")
```

```
cat(sprintf("Slope: %.6f | SE: %.6f | t: %.4f | p: %s\n95%% CI: [%.6f, %.6f]\n",
          m1$beta[2], m1$se, m1$t, format.pval(m1$p),
          m1$ci_beta[1], m1$ci_beta[2]))

cat("\n")
cat(loc_name, "|", late_label, "\n")
cat(sprintf("Slope: %.6f | SE: %.6f | t: %.4f | p: %s\n95%% CI: [%.6f, %.6f]\n",
          m2$beta[2], m2$se, m2$t, format.pval(m2$p),
          m2$ci_beta[1], m2$ci_beta[2]))

plot(year, y, pch = 16, col = "grey40",
    xlab = "Year", ylab = "Temperature (°C)",
    main = paste(dataset_name, "-", loc_name),
    cex.main = 1.1)

lines(year[early_idx], m1$yhat,  col = "blue", lwd = 2)
lines(year[early_idx], m1$upper, col = "blue", lty = 2)
lines(year[early_idx], m1$lower, col = "blue", lty = 2)

lines(year[late_idx],  m2$yhat,  col = "red",  lwd = 2)
lines(year[late_idx],  m2$upper, col = "red",  lty = 2)
lines(year[late_idx],  m2$lower, col = "red",  lty = 2)

legend("topleft",
      legend = c("Observed",
             paste("Trend", early_label),
             paste("CI", early_label),
```

```
              paste("Trend", late_label),
              paste("CI", late_label)),
         col = c("grey40","blue","blue","red","red"),
         pch = c(16,NA,NA,NA,NA),
         lty = c(NA,1,2,1,2),
         lwd = c(NA,2,1,2,1),
         bty = "n", cex = 0.7)
  }

  par(mfrow = c(1,1))
}
#load annual data
data_annual <- read.csv("AnnualTemp.csv", sep=";", dec=",")
colnames(data_annual) <- c("Date", "De.Bilt", "Eelde", "Maastricht")

#load monthly data
data_monthly <- read.csv("SmoothedMonthlyTemp.csv", sep=";", dec=",")
colnames(data_monthly) <- c("Date", "De.Bilt", "Eelde", "Maastricht")

#run analysis
run_temperature_analysis(data_annual,  "Annual Data",
                split_year = 1960,
                end_year = 2024)

run_temperature_analysis(data_monthly, "Smoothed Monthly Data",
                split_year = 1960,
                end_year = 2024)
```

Appendix 7: Bootstrap code

```
rm(list=ls())

# OLS

ols_full <- function(x, y) {


 n <- length(y)

 x_bar <- mean(x)

 y_bar <- mean(y)


 SXX <- sum((x - x_bar)^2)

 SXY <- sum((x - x_bar) * (y - y_bar))


 beta  <- SXY / SXX

 alpha <- y_bar - beta * x_bar


 y_hat <- alpha + beta * x

 e    <- y - y_hat


 sigma2 <- sum(e^2) / (n - 2)


 se_beta  <- sqrt(sigma2 / SXX)

 se_alpha <- sqrt(sigma2 * (1/n + x_bar^2 / SXX))


 Q <- beta / se_beta


 list(alpha = alpha,

    beta  = beta,
```

```
      se_beta  = se_beta,
      se_alpha = se_alpha,
      Q = Q)
}
# time variable
create_time <- function(year) {
  if (max(year) > 10000) {
    return((1:length(year)) / 12)
  } else {
    return(1:length(year))
  }
}

# BOOTSTRAP (centered Q*)
bootstrap_paired <- function(x, y, B = 2000, alpha = 0.05) {

  n <- length(y)
  orig <- ols_full(x, y)

  Q_star <- numeric(B)
  beta_star <- numeric(B)
  alpha_star <- numeric(B)

  for (b in 1:B) {

    J <- sample(1:n, n, replace = TRUE)
```

```
  x_b <- x[J]
  y_b <- y[J]

  m_b <- ols_full(x_b, y_b)

  Q_star[b] <- (m_b$beta - orig$beta) / m_b$se_beta
  beta_star[b]  <- m_b$beta
  alpha_star[b] <- m_b$alpha
}

crit <- quantile(Q_star, 1 - alpha)
p_val <- mean(Q_star >= orig$Q)

ci_beta  <- quantile(beta_star,  c(alpha/2, 1-alpha/2))
ci_alpha <- quantile(alpha_star, c(alpha/2, 1-alpha/2))

list(
  Q = orig$Q,
  crit = crit,
  p = p_val,

  alpha = orig$alpha,
  beta = orig$beta,

  alpha_star = mean(alpha_star),
  beta_star  = mean(beta_star),
```

```
    ci_beta = ci_beta,
    ci_alpha = ci_alpha,


    se_beta_star  = sd(beta_star),
    se_alpha_star = sd(alpha_star),


    Q_star = Q_star
  )
}
# main function
run_bootstrap_all_cities <- function(data, dataset_name, split_year = 1960) {


  cities <- c("Maastricht", "De.Bilt", "Eelde")
  year <- data$Date


  if (max(year) > 10000) {
    year_numeric <- floor(year / 100)
  } else {
    year_numeric <- year
  }


  cat("\n=================================\n")
  cat("BOOTSTRAP:", dataset_name, "\n")
  cat("=================================\n")


  for (city in cities) {
```

```
y <- data[[city]]

early_idx <- year_numeric <= split_year
late_idx  <- year_numeric > split_year

x1 <- create_time(year_numeric[early_idx])
y1 <- y[early_idx]

x2 <- create_time(year_numeric[late_idx])
y2 <- y[late_idx]

res1 <- bootstrap_paired(x1, y1)
res2 <- bootstrap_paired(x2, y2)

# Print
cat("\n---------------------------------\n")
cat("City:", city, "\n")
cat("---------------------------------\n")

cat("EARLY PERIOD\n")
cat(sprintf("beta = %.6f | beta_star = %.6f\n", res1$beta, res1$beta_star))
cat(sprintf("Q = %.4f | crit = %.4f | p = %.6f\n",
        res1$Q, res1$crit, res1$p))
cat(sprintf("CI_beta = [%.6f , %.6f]\n",
        res1$ci_beta[1], res1$ci_beta[2]))


cat("\nLATE PERIOD\n")
```

```
cat(sprintf("beta = %.6f | beta_star = %.6f\n", res2$beta, res2$beta_star))
cat(sprintf("Q = %.4f | crit = %.4f | p = %.6f\n",
          res2$Q, res2$crit, res2$p))
cat(sprintf("CI_beta = [%.6f , %.6f]\n",
          res2$ci_beta[1], res2$ci_beta[2]))

# PLOTS (FOR EACH CITY)
par(mfrow = c(1, 2))
# EARLY
x_min1 <- min(res1$Q_star, res1$Q, res1$crit)
x_max1 <- max(res1$Q_star, res1$Q, res1$crit)

hist(res1$Q_star,
    breaks = 30,
    col = "lightblue",
    main = paste(dataset_name, "-", city, "- Early (1907–1960)"),
    xlab = "Q statistic",
    xlim = c(x_min1, x_max1))

abline(v = res1$Q, col = "red", lwd = 3)
abline(v = res1$crit, col = "darkgreen", lwd = 3)

legend("topright",
      legend = c("Observed Q", "Critical value"),
      col = c("red", "darkgreen"),
      lwd = 3)
# LATE
```

```r
    x_min2 <- min(res2$Q_star, res2$Q, res2$crit)
    x_max2 <- max(res2$Q_star, res2$Q, res2$crit)
    hist(res2$Q_star,
         breaks = 30,
         col = "lightblue",
         main = paste(dataset_name, "-", city, "- Late (1961–2024)"),
         xlab = "Q statistic",
         xlim = c(x_min2, x_max2))

    abline(v = res2$Q, col = "red", lwd = 3)
    abline(v = res2$crit, col = "darkgreen", lwd = 3)
    legend("topright",
           legend = c("Observed Q", "Critical value"),
           col = c("red", "darkgreen"),
           lwd = 3)

    par(mfrow = c(1,1))
  }
}
# LOAD DATA
data_annual <- read.csv("AnnualTemp.csv", sep=";", dec=",")
colnames(data_annual) <- c("Date", "De.Bilt", "Eelde", "Maastricht")
data_monthly <- read.csv("SmoothedMonthlyTemp.csv", sep=";", dec=",")
colnames(data_monthly) <- c("Date", "De.Bilt", "Eelde", "Maastricht")
# RUN
run_bootstrap_all_cities(data_annual,  "Annual Data")
run_bootstrap_all_cities(data_monthly, "Smoothed Monthly Data")
```